



The ABC transporter, AbcB3, mediates cAMP export in *D. discoideum* development

Edward Roshan Miranda^{a,b,1}, Edward A. Nam^{b,1}, Adam Kuspa^{b,c}, Gad Shaulsky^{b,*}

^a Graduate Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030, USA

^b Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030, USA

^c Department of Biochemistry and Molecular Biology, Baylor College of Medicine, Houston, TX 77030, USA

ARTICLE INFO

Article history:

Received 22 July 2014

Received in revised form

15 October 2014

Accepted 11 November 2014

Available online 20 November 2014

Keywords:

ATP-binding cassette transporters

Dictyostelium

cAMP

ABSTRACT

Extracellular cAMP functions as a primary ligand for cell surface cAMP receptors throughout *Dictyostelium discoideum* development, controlling chemotaxis and morphogenesis. The developmental consequences of cAMP signaling and the metabolism of cAMP have been studied in great detail, but it has been unclear how cells export cAMP across the plasma membrane. Here we show pharmacologically and genetically that ABC transporters mediate cAMP export. Using an evolutionary-developmental biology approach, we identified several candidate *abc* genes and characterized one of them, *abcB3*, in more detail. Genetic and biochemical evidence suggest that AbcB3 is a component of the cAMP export mechanism in *D. discoideum* development.

© 2014 Elsevier Inc. All rights reserved.

Introduction

When starved, the single-celled social amoeba *Dictyostelium discoideum* undergoes a remarkable developmental process that includes chemotaxis, aggregation, kin recognition, and morphogenesis, resulting in a multicellular fruiting body (Kessin, 2001). Over 40 years ago, cAMP was identified as the chemotactic signal for aggregation (Barkley, 1969). Extracellular cAMP also functions as a first messenger to specify cell fate during development (Alvarez-Curto et al., 2005; Schaap, 2011; Saran and Schaap, 2004; Virdy et al., 1999). Despite the developmental importance of extracellular cAMP, how cAMP export occurs has remained unknown.

Several groups have proposed mechanisms for cAMP export, including a vesicular transport mechanism. However, cAMP export occurs simultaneously with intracellular synthesis with virtually no lag time during the initial stages of development (Dinauer et al., 1980), suggesting that vesicular transport is not at play. Adenylyl cyclase A (ACA) synthesizes intracellular cAMP and contains a structure similar to ATP Binding Cassette (ABC) transporters (Kessin, 2001; Pitt et al., 1992), which are known to export anionic cargoes like cAMP in other organisms (Cheepala et al., 2013). While ACA is an attractive candidate for cAMP export, *acaA*-null mutants can export cAMP when another adenylyl cyclase, ACG, is engineered to produce intracellular cAMP, indicating that ACA is not the only

possible cAMP exporter (Pitt et al., 1993). Interestingly, several human ABC transporters can export cAMP (Chen et al., 2001; van Aubel et al., 2002; Wielinga et al., 2003), prompting us to hypothesize that ABC transporters export cAMP in *Dictyostelium*.

ABC transporters contain two ATP-binding domains and two membrane-spanning domains that transport a variety of cargo molecules (Cheepala et al., 2013; Anjard and Loomis, 2002; Annilo et al., 2006; Graf et al., 2003; Sarkadi et al., 2006). In *D. discoideum* there are 68 *abc* genes, which have been classified into 8 groups (*abcA* through *abcH*) based on sequence and predicted structural features (Anjard and Loomis, 2002), but their functions are largely unknown. We recently characterized the transcriptional and morphological phenotypes of 35 *abc* mutants during development and found that most mutants exhibited subtle phenotypes, suggesting that many of the ABC transporters could have overlapping functions (Miranda et al., 2013). We hypothesized that cAMP exporters may be among the uncharacterized *abc* genes.

Here we show that ABC transporter inhibitors disrupt development in a manner consistent with a lack of cAMP export. We also identify several *abc* genes with cAMP-responsive transcription in *D. discoideum* and in *Dictyostelium purpureum*. Using genetic and biochemical approaches, we show that one of these genes, *abcB3*, exports cAMP.

Results

ABC-transporter inhibitors disrupt *D. discoideum* development

Bacterial and human ABC transporters that export similar cargo molecules exhibit overlap in inhibitor sensitivity (van Veen et al.,

* Corresponding author.

E-mail address: gadi@bcm.edu (G. Shaulsky).

¹ Equal contribution.

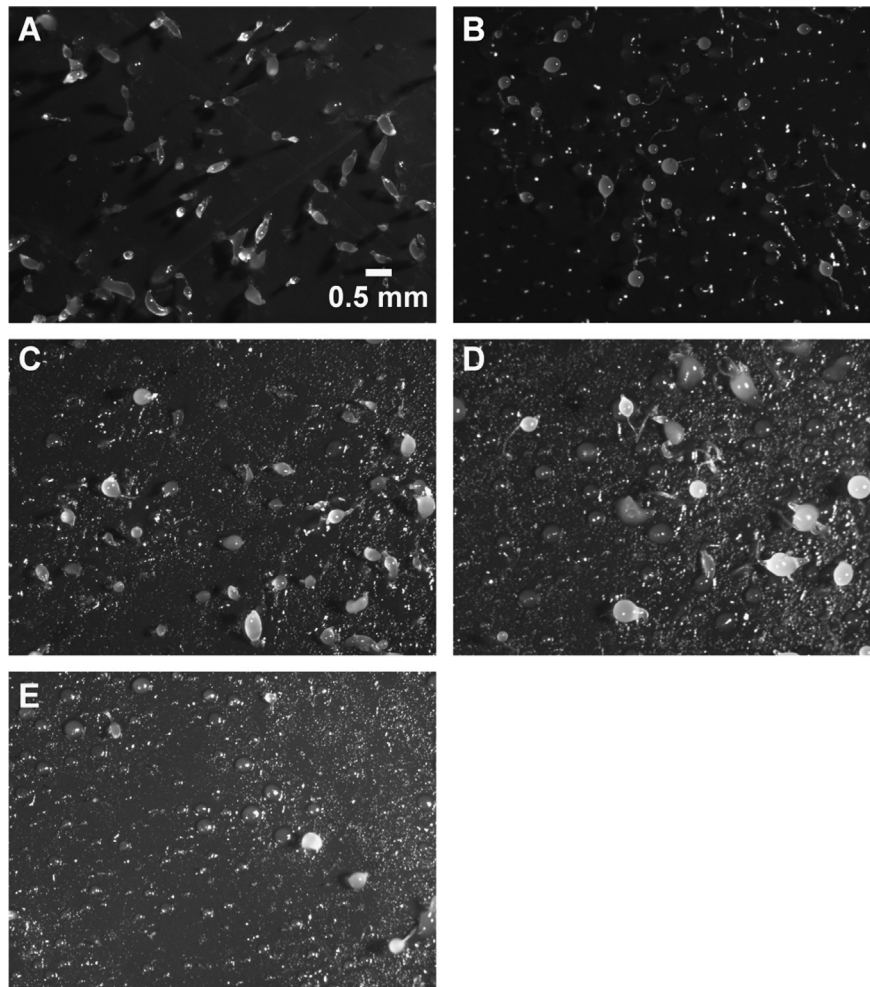


Fig. 1. ABC transporter inhibitors disrupt *D. discoideum* development. Cells were developed on black filters in the presence of (A) no drug, (B) dipyridamole, (C) indomethacin, (D) probenecid, and (E) sulfapyrazone. Images shown were taken 22 h into development and are representative images of 3 independent experiments.

1998), suggesting that pharmacological properties of ABC transporters may be similar when they share cargo molecules. The drugs Indomethacin, Probenecid, Dipyridamole, and Sulfapyrazone inhibit the human cAMP exporters, ABCC4 and ABCC5 (Wielinga et al., 2003; Reid et al., 2003). To test whether ABC transporters export cAMP in *D. discoideum*, we examined the effect of these inhibitors on development. Development of wild-type *D. discoideum* (AX4) cells without inhibitors for 22 h resulted in synchronous development and the formation of culminants (Fig. 1A). Addition of inhibitors disrupted development and caused asynchronous morphogenesis (Fig. 1B–E). Sulfapyrazone caused the earliest arrest at the loose aggregate stage (Fig. 1E) and dipyridamole treatment had the least effect, resulting in semi-clear, frail fruiting bodies (Fig. 1B). Transferring cells that were initially developed on inhibitor-containing filters to inhibitor-free conditions reversed the developmental arrests, and the cells developed into fruiting bodies after another 24 h. The reversibility of the drug effects indicates that the treatment was not lethal.

Before individual cells form multicellular structures, they stream towards aggregation centers in an extracellular cAMP-dependent manner (Kessin, 2001). We tested the effect of ABC-transporter inhibitors on the streaming of developing cells in submerged cultures. In the absence of inhibitor, we observed aggregation centers with radial tracks of cells aggregating towards the center (Fig. 2A). Addition of any one of the inhibitors abolished developmental streaming and aggregation (Fig. 2B–E). In the previous experiment we used a high concentration of inhibitors (Fig. 1)

because the drugs were added to buffer-soaked paper pads beneath the developmental filters. However, inhibition of developmental streaming in submerged cultures occurred at drug concentrations of 50 μ M, similar to the concentrations used in human cell culture experiments (Wielinga et al., 2003; Reid et al., 2003). These results demonstrate that inhibitors of ABC transporters disrupt cAMP-mediated development and support the hypothesis that ABC transporters export cAMP in *D. discoideum*.

Several *abc* genes are cAMP responsive

To identify candidate cAMP exporters among the 68 ABC transporter genes in *D. discoideum* we relied on the observation that excess cargo molecules induce the transcription of their respective ABC transporters in other organisms (Venkateswaran et al., 2000a, 2000b; Zhou et al., 2008). We developed *D. discoideum* cells for 0, 3, and 6 h and treated them with the membrane permeable cAMP analog, 8-Br-cAMP, for 2 h to mimic intracellular cAMP cargo overload (Riley and Barclay, 1990). We then measured the transcript abundance of each of the 68 ABC transporters by quantitative RT-PCR (qRT-PCR) and compared them between treated and untreated cells. We found that *abc*-transcript abundance was not altered at 0 and 3 h, but *abcA5*, *abcB3*, *abcC5*, *abcC12*, *abcC13*, and *abcG21* transcripts accumulated in cells developed for 6 h in the presence of 8-Br-cAMP (Fig. 3A).

To further refine our candidate gene list, we took a comparative evolutionary-developmental biology approach. Similar to *D. discoideum*, *D. purpureum* utilizes cAMP signaling for chemotactic

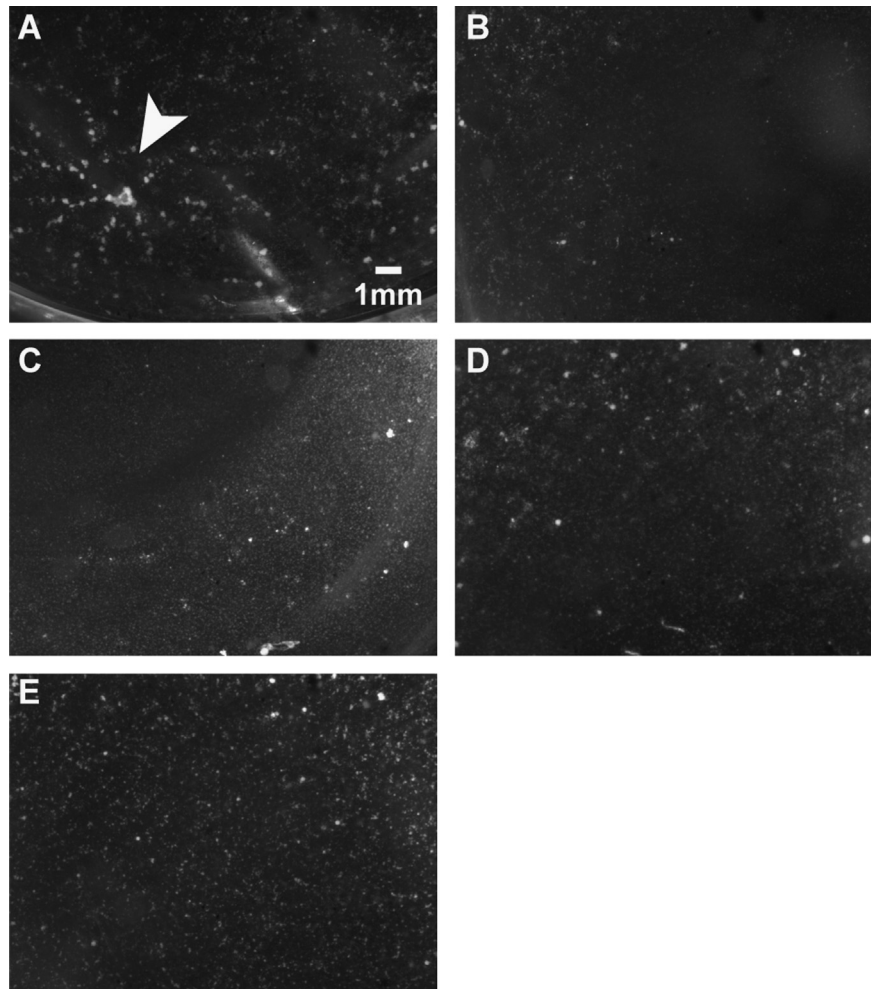


Fig. 2. ABC transporter inhibitors prevent *D. discoideum* developmental streaming. Cells were developed for 16 h in submerged cultures in the presence of (A) no drug, (B) dipyrindamole, (C) indomethacin, (D) probenecid, and (E) sulfapyrazone. White arrow indicates the wild type pattern of cells streaming towards an aggregation center. Images are representative of 3 independent experiments.

aggregation. However, these two species diverged about 300–450 million years ago and possess divergent genomes (Schaap et al., 2006; Sucgang et al., 2013). We hypothesized that *abc* genes responsible for cAMP export would be shared between the two species. To test this hypothesis, we repeated the cargo overload experiments with *D. purpureum* and measured transcript abundance of the orthologs of the *abc* genes up-regulated in *D. discoideum*. The qRT-PCR results showed that *abcA5*, *abcB3*, and *abcG21* transcript abundance increased with the addition of 8-Br-cAMP, whereas *abcC5* and *abcC12* did not (Fig. 3B). There is no obvious ortholog for *abcC13* in the *D. purpureum* genome. These findings suggest that *abcA5*, *abcB3*, and *abcG21* encode cAMP exporters, since cargo overload up-regulated the respective transcripts in two evolutionary divergent organisms that use cAMP for chemotaxis.

The cAMP analog 8-Br-cAMP is a potent activator of Protein Kinase A (PKA) (Richardson et al., 1994), and at least one *abc* gene, *abcA1*, is up-regulated through the PKA pathway in another system even though cAMP is not a cargo molecule for this transporter (Oram et al., 2000). To distinguish between PKA-regulated and cAMP cargo-regulated *abc* transcripts, we analyzed transcript levels in two mutants with constitutively active forms of PKA in *D. discoideum*. In cells overexpressing the PKA catalytic subunit (*pkaC^{OE}*), *abcC13* and *abcG21* transcript abundance was increased more than twofold compared to the wild type (Fig. 3C). Knockout of the gene that encodes the inhibitory regulatory

subunit of PKA, *pkaR*, caused an increased in *abcA5* and *abcG21* transcripts (Fig. 3D). These results implicate *abcB3* as the most likely candidate because its expression was up-regulated in response to cargo overload in an evolutionarily conserved manner (Fig. 3A and B), and because it was not regulated by PKA (Fig. 3C and D).

abcB3 knockdown delays *D. discoideum* development

We hypothesized that *abcB3* exports cAMP during *D. discoideum* development and predicted that knockout of the gene would result in developmental defects, similar to those seen with the inhibitor experiments (Fig. 1). However, we were unable to generate a knockout strain, suggesting that *abcB3* might be required for vegetative growth. As an alternative approach, we created *abcB3*-RNAi strains. During development, *abcB3* expression increases at 4 h and peaks by 8 h (Miranda et al., 2013; Parikh et al., 2010), so we drove expression of the *abcB3* shRNA using the *tgrB1* element of the bidirectional *tgrB1/C1* promoter (Benabentos et al., 2009; Hirose et al., 2011). Since *tgrB1* expression begins at the onset of development, peaks at 4 h, and remains high until 8 h, we reasoned that the construct would allow for high levels of shRNA to be present when *abcB3* expression increases and peaks, but not during growth. The bidirectional promoter also allowed us to co-express GFP from the *tgrC1* element, confirming expression of the *abcB3* shRNA by visualizing GFP positive cells during development. After 10 h of development, wild-type cells

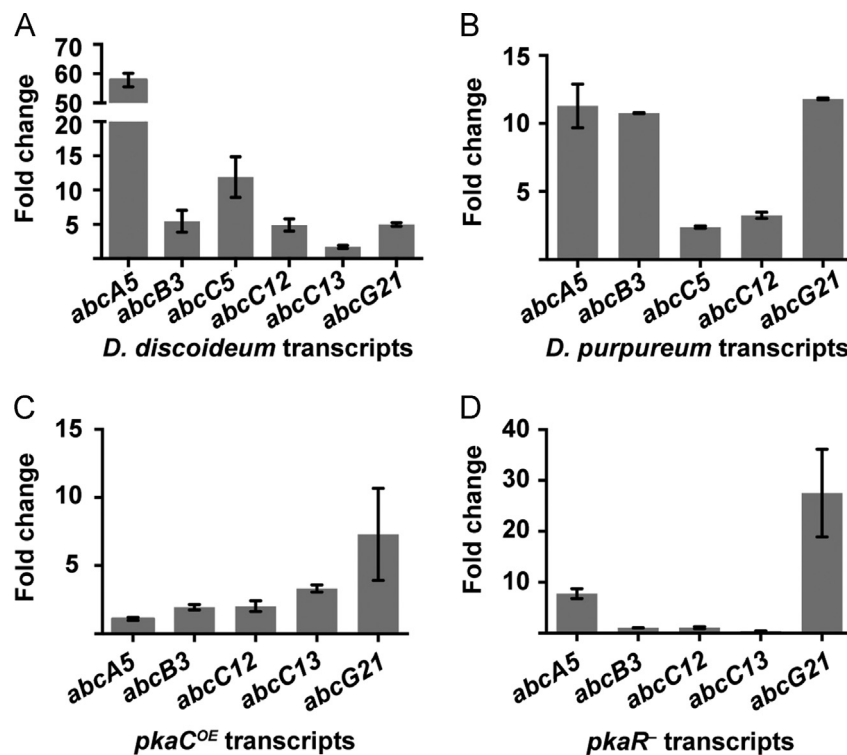


Fig. 3. Cargo overload increases *abc* transporter mRNA abundance. (A) *D. discoideum* and (B) *D. purpureum* cells were developed on black filters and treated with 8-Br-cAMP. Six hours after treatment, we measured transcript levels for the 68 *abc* transporter genes by qRT-PCR. Fold change (y-axis) was calculated as the ratio of the mRNA abundance between the treated and the untreated samples. Genes showing more than a two fold increase in abundance are shown. (C) *pkaC^{OE}* and (D) *pkaR⁻* mutant *D. discoideum* cells were developed for 6 h and transcript levels of selected *abc* genes from A were measured by qRT-PCR. Fold increase was calculated as the ratio of the mRNA abundance between the mutant and the wild type. Bars represent the means and error bars represent standard error of the mean of three independent replications. Note that the scales of the y-axes are different between the different panels.

showed low background levels of green fluorescence (Fig. 4A), whereas two independent strains transformed with the *abcB3* shRNA construct showed strong GFP expression (Fig. 4A). To test *abcB3* mRNA abundance in these cells, we measured mRNA levels at 10 h of development by qRT-PCR and found that *abcB3* expression was significantly reduced in both strains compared to the wild type (Fig. 4B). The level of knockdown was different between the two *abcB3*-RNAi strains, so we used both strains in subsequent experiments.

To test the effects of reduced *abcB3* expression, we followed the wild type and both *abcB3*-RNAi strains during development. At 6 h, the wild type formed loose aggregates while strain *abcB3*-RNAi #1 began to form loose aggregates. Strain *abcB3*-RNAi #2, however, failed to form loose aggregates by 6 h (Fig. 5). These results are consistent with the greater reduction of *abcB3* mRNA in strain #2. At 10 h, the developmental differences between the wild type and the *abcB3* RNAi strains were clearest. The wild-type cells formed tipped aggregates while *abcB3*-RNAi #1 and #2 cells remained at the tight aggregate and loose aggregate stages, respectively (Fig. 5). The developmental delays persisted at 17, 22, and 27 h. By 27 h the majority of the wild-type cells formed fruiting bodies while the *abcB3*-RNAi strains did not (Fig. 5). The developmental delay displayed by the *abcB3*-RNAi cells is consistent with reduced cAMP export.

abcB3 knockdown reduces extracellular cAMP levels

To test whether *abcB3*-RNAi expression resulted in altered cAMP levels, we developed cells for 6 h, collected them and stimulated cAMP production with the cAMP analog 2'-deoxy-cAMP. This analog binds to cell surface cAMP receptors and induces cAMP production and export, but it is not detected by the cAMP detection method we

used (Van Haastert, 2006; Chen et al., 2002). We found that stimulation caused a twofold increase in the levels of extracellular cAMP in the wild type but no significant increase in the two *abcB3*-RNAi cells (Fig. 6A). This finding supports the hypothesis that AbcB3 exports cAMP during development. In wild-type cells, intracellular cAMP levels did not significantly differ between unstimulated and stimulated conditions (Fig. 6B), probably because most of the newly synthesized cAMP was exported (Fig. 6A). We observed that intracellular cAMP levels did not increase significantly in *abcB3*-RNAi cells either (Fig. 6B). We suspect that reduced cAMP export could cause reduced cAMP production because cAMP levels are tightly regulated through feedback mechanisms (Saran et al., 2002).

AbcB3 is an exporter of cAMP

To examine the cAMP-exporter activity of AbcB3 more directly, we tested whether *abcB3* expression in an ectopic system could facilitate cAMP export. We transiently transfected human HEK293T cells with *abcB3* cDNA or with an empty vector (pcDNA3) control, stimulated the endogenous human adenylyl cyclase with the addition of Forskolin, and measured cAMP levels. In vector control cells, Forskolin addition caused an increase in extracellular cAMP levels (Fig. 7A). Ectopic expression of AbcB3 caused an additional 30% increase in extracellular cAMP accumulation with Forskolin addition (Fig. 7A; black bars). Moreover, addition of the ABC-transporter inhibitor Probenicid reduced extracellular cAMP levels in AbcB3 transfected cells to that of the control cells (Fig. 7A; gray bars). This increase in cAMP export is accompanied by a decrease in intracellular cAMP levels after Forskolin treatment in AbcB3 expressing cells compared to control cells (Fig. 7B; black bars). Consistent with other studies in mammalian cells, inhibitor treatment did not raise intracellular cAMP levels likely due to

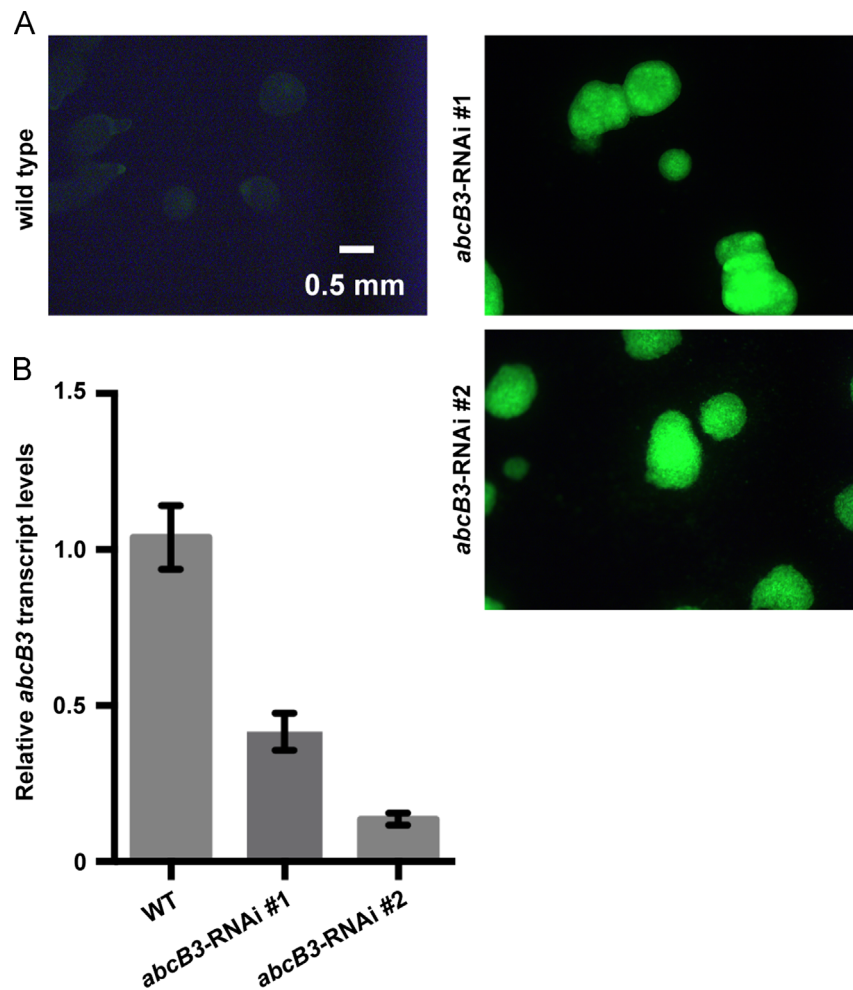


Fig. 4. Expression of *abcB3* shRNA reduces *abcB3* transcripts. Wild-type and *abcB3* shRNA expressing cells were developed on black filters for 10 h as indicated. (A) GFP expression, an indicator of transformation efficiency, was determined by fluorescence microscopy. (B) *abcB3* transcript levels were measured by qRT-PCR and calculated as the ratio of the mRNA abundance between the mutant cells and the wild type (y-axis). Bars represent the means and error bars represent standard error of the mean of three independent replications.

negative feedback regulation of cAMP production (Wielinga et al., 2003; Ardaillou et al., 1993; Hamet et al., 1989; Patel et al., 1995; Tjornhammar et al., 1983; Heasley and Brunton, 1985). These data further support the hypothesis that AbcB3 can export cAMP.

Discussion

Our data suggest that AbcB3 functions as a cAMP exporter in *Dictyostelium*. *abcB3* mRNA accumulates in response to cAMP overload in *D. discoideum* and in the evolutionary distant relative, *D. purpureum*, which also uses cAMP as an aggregation chemoattractant. RNAi-mediated knockdown of *abcB3* results in decreased extracellular cAMP levels and leads to perturbed *D. discoideum* development and ectopic expression of *abcB3* is sufficient to increase cAMP export in mammalian cells.

Pharmacological inhibition of ABC transporters disrupted developmental aggregation and caused developmental delay, suggesting that ABC transporters are involved in cAMP export. The ABC transporter inhibitors used are chemically distinct and their potential off-target effects are probably different. However, all inhibitors disrupted development similarly and in a manner consistent with a defect in cAMP export (Wang and Kuspa, 1997), suggesting that their common effects are due to reduced cAMP export.

In our genetics studies we used three criteria to select *abc* genes for further analysis: (1) cAMP-induced transcriptional regulation, (2) PKA-independent transcriptional regulation, and (3) evolutionary conservation of cAMP-dependent transcriptional regulation. Remarkably, of the 68 *abc* genes, *abcB3* was the only gene that satisfied all three criteria. While our data suggests that *abcB3* is indeed a cAMP exporter, we have not excluded the possibility that other cAMP-responsive *abc* genes encode potential cAMP exporters.

In gene silencing experiments, we observed a developmental delay that began at the aggregation stage. The initiation of the delay at the aggregation stage is in line with a defect in cAMP signaling (Kessin, 2001; Saran et al., 2002; Meima and Schaap, 1999). The delay persisted throughout development; however, some cells were able to form fruiting bodies eventually. The lack of complete developmental arrest suggests that there are other cAMP exporters in *D. discoideum*. This would not be surprising considering that ABC proteins are promiscuous transporters (Cheepala et al., 2013; Anjard and Loomis, 2002; Annilo et al., 2006; Sarkadi et al., 2006; Dean et al., 2001; Frelet and Klein, 2006; Linton, 2007; van de Water et al., 2005). A more interesting rationale for multiple cAMP exporters is that *D. discoideum* development is controlled by several cAMP receptors (CAR1–4) expressed at different times and in specific cell types with varying degrees of affinity to cAMP (Prabhu and Eichinger, 2006; Saxe et al., 1991). It is conceivable that multiple cAMP exporters exist

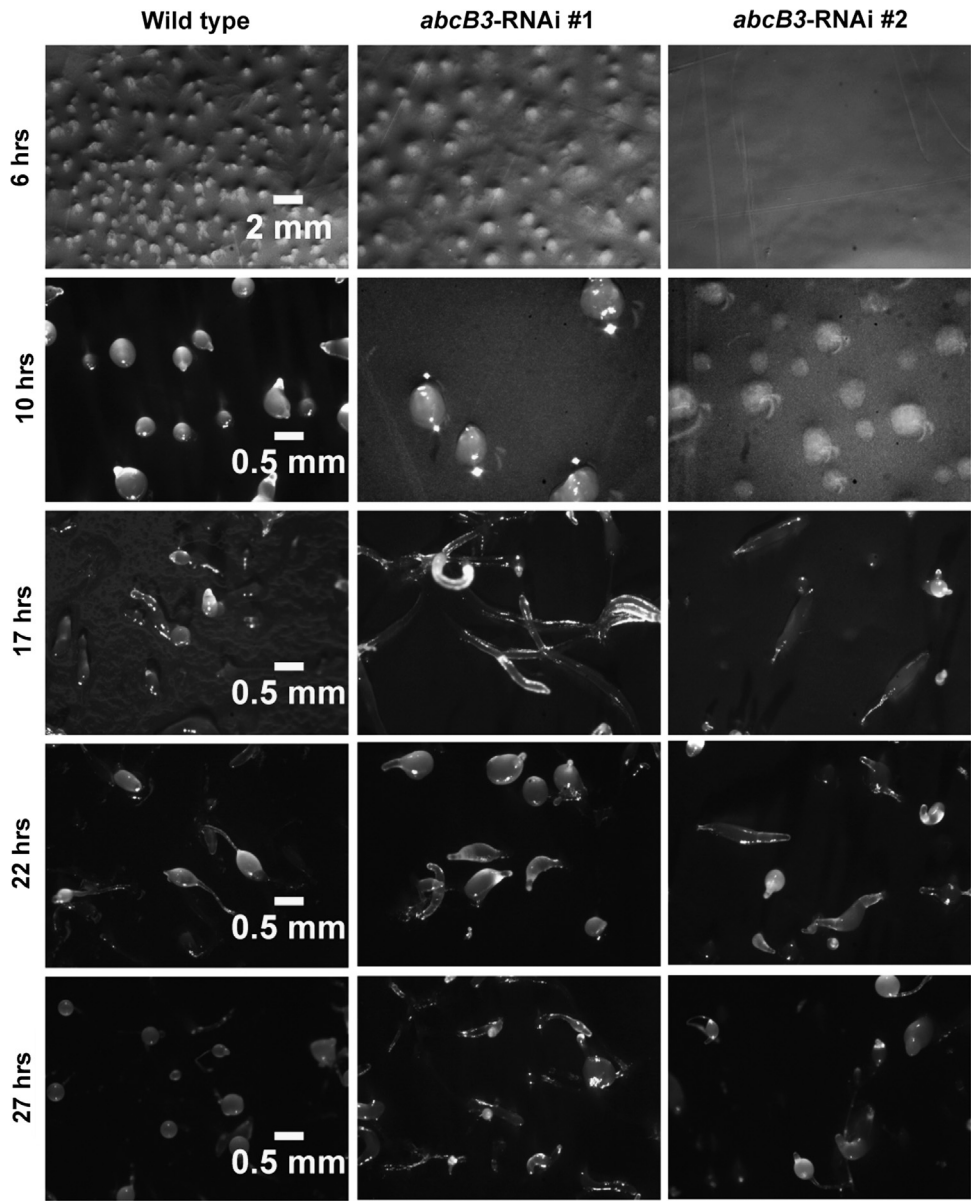


Fig. 5. *abcB3* knockdown delays *D. discoideum* development. Wild-type and *abcB3* shRNA expressing cells were developed on black filters and images taken at the indicated times. Images are representative of results from 3 independent experiments. Two independent strains expressing *abcB3* shRNA constructs are shown. Scale bars for each row are shown in the left panels.

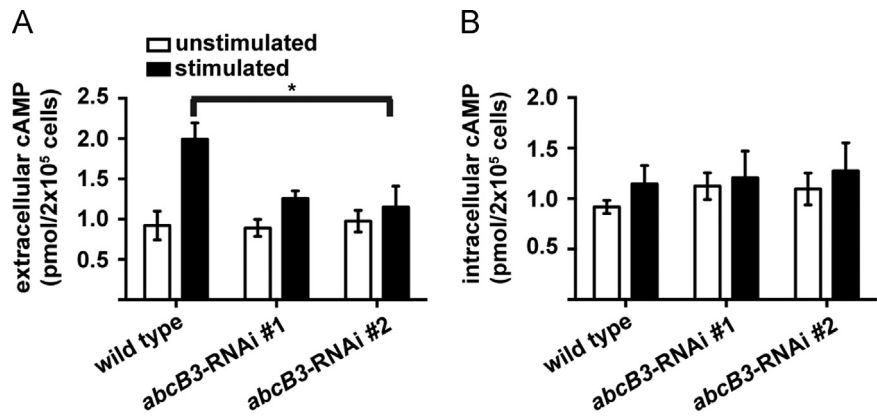


Fig. 6. *abcB3* knockdown reduces extracellular levels of cAMP in *D. discoideum*. Wild-type and *abcB3* shRNA expressing cells were developed on black filters for 6 h, disaggregated, and treated with 2'-deoxy-cAMP. We measured extracellular (A) and intracellular (B) cAMP levels (pmol/2 × 10⁵ cells). Bars represent the means and error bars represent standard error of the mean of four independent replications. **p* ≤ 0.02.

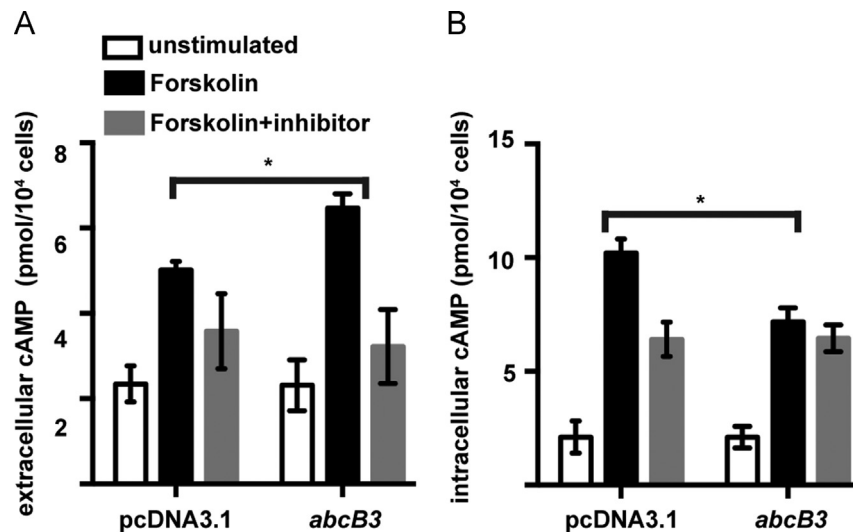


Fig. 7. Ectopically expressed *abcB3* exports cAMP in mammalian cells. HEK293T cells were transiently transfected with pcDNA3.1 (empty vector control) or an expression vector for *abcB3* cDNA, as indicated. Transfected cells were unstimulated, stimulated with Forskolin alone, or with Forskolin and the ABC transporter inhibitor Probenecid, as indicated. We measured extracellular (A) and intracellular (B) cAMP levels (pmol/10⁴ cells). Bars represent the means and error bars represent standard error of the mean of three independent replications. **p* < 0.02.

and are also differentially expressed temporally and spatially during multicellular development. The other cAMP-responsive *abc* genes we identified might compensate when *abcB3* is down-regulated. Additionally, it is still formally possible that Adenylyl Cyclase A exports cAMP because the experiments done to test its potential export activity only showed that it was not the only exporter (Pitt et al., 1993).

Another explanation for a lack of developmental arrest with *abcB3*-RNAi is that *abcB3* knockdown was transient and incomplete. We were unable to knockout the *abcB3* gene, which suggests that it might have an additional role in *D. discoideum* growth. Instead, we used the *tgrB1-C1* promoter to drive *abcB3*-RNAi expression. The *tgrB1-C1* promoter becomes active at around 4 h of development in all cells and remains active for about 6 h, after which it is mostly active in prestalk cells (Parikh et al., 2010; Benabentos et al., 2009; Hirose et al., 2011; Loomis and Shaulsky, 2011). It is possible that once the *tgrB1-C1* promoter became inactive, the levels of *abcB3*-RNAi expression were insufficient to reduce the accumulation of *abcB3* mRNA, and the cells were able to recover and develop. The reversibility of the phenotype is also supported by the observation that the ABC-transporter inhibitors effects were reversible. Both insufficient knockdown and the presence of multiple cAMP exporters likely contribute to the developmental phenotype observed in our experiments.

We measured cAMP levels inside and outside of *D. discoideum* cells and found a positive relationship between *AbcB3* expression and levels of extracellular cAMP. We do not have direct binding evidence or structural data to further support that *AbcB3* exports cAMP. It is therefore formally possible that *AbcB3* may function in some ancillary capacity to the real cAMP exporter. However, the most compelling evidence that *AbcB3* exports cAMP is that expressing *abcB3* in mammalian cells was sufficient to increase extracellular cAMP levels without any additional *D. discoideum* genes. Moreover, ABC transporter inhibitors blocked that increase in extracellular export.

ABC transporters are present in every domain of life, transport an incredible array of molecules, and show a considerable level of functional and structural homology from bacteria to vertebrates. Human ABC transporters can export cAMP, and our data suggest that this function is also conserved in *Dictyostelium*. The physiological significance of cAMP export is clear in *D. discoideum* (Alvarez-Curto et al., 2005; Prabhu and Eichinger, 2006). Very

little is known about the function of cAMP efflux in mammals (Hamet et al., 1989; Tjornhammar et al., 1983), but the fact that it exists suggests that the ability to export cAMP evolved before the split between animals and amoebae.

Much remains to be examined with regards to cAMP export and *AbcB3* function and regulation. From a broader perspective, there are many *abc* genes in the *D. discoideum* genome, but their cargoes and functions are largely unknown. Multidrug resistance caused by ABC transporters has been an area of focused research and ABC transporter dysfunction causes 17 known genetic disorders (Annino et al., 2006). Our previous work (Miranda et al., 2013) and this study provide a framework for studying ABC transporter function in *D. discoideum* not only in the context of amoeba biology but also in the context of human biology and disease.

Methods

Dictyostelid strains, growth, and development

D. discoideum AX4 (wild type), *pkaR*[−], and *pkaC*^{OE} strains are described in Shaulsky et al. (1998), Knecht et al. (1986), Anjard et al. (1992) and Mann et al. (1992). *D. purpureum* DpAX1 is described in Sugang et al. (2011). Cells were grown in shaking suspension at 220 rpm at 22°C in HL5 medium supplemented with penicillin (50 U/mL) and streptomycin (300 µg/mL) (Sussman, 1987). *abcB3* shRNA expressing cells were maintained as above and supplemented with G418 (10 µg/mL). For development on black nitrocellulose filters, cells were harvested at the logarithmic growth phase and washed twice with KK2 buffer (16.3 mM KH₂PO₄, 3.7 mM K₂HPO₄, pH 6.2). Cells (5 × 10⁷) were deposited onto a 5 cm filter atop a paper pad saturated with 2 mL PDF buffer (9.2 mM K₂HPO₄, 13.2 mM KH₂PO₄, 20 mM KCl, 1.2 mM MgSO₄, pH 6.5) and incubated in a dark humid chamber at 22°C. Where indicated, vehicle control or 1 mM of the ABC inhibitors Indomethacin, Probenecid, Dipyrindamole, or Sulfynpyrazone (Sigma-aldrich, MO, USA) was added to the PDF buffer. For developmental streaming assays, 1.56 × 10⁶ cells/cm² were deposited in a 12 well plate (BD Biosciences, CA, USA) under 666 µL of PDF containing vehicle control or 50 µM of one of the ABC transporter inhibitors. Cells were incubated at 22°C in a dark chamber and imaged at 16 h.

Plasmids, transformations, and transfections

The *abcB3* RNAi plasmid was constructed as described in (Martens et al., 2002). Briefly, bases 80–480 of the *abcB3* coding sequence followed by sequences complementary to bases 302–480 were subcloned into an expression vector under control of the bidirectional *tgrB1/tgrC1* promoter to facilitate RNAi during development. The final vector contained *abcB3* sequences controlled by the *tgrB1* promoter and a green fluorescence protein (GFP) gene controlled by the *tgrC1* promoter, resulting in co-expression of *abcB3* RNAi sequences and GFP. This plasmid was transformed into the AX4 strain, and G418 resistant clones were selected as described in (Benabentos et al., 2009). Independent strains were isolated from independent transformations. *abcB3* knockdown was confirmed by quantitative RT-PCR (qRT-PCR).

abcB3 cDNA was cloned in the pcDNA3.1(+) plasmid (Invitrogen, CA, USA). HEK293T cells (Nam et al., 2011a, 2011b) were maintained in DMEM (Invitrogen) containing 10% fetal bovine serum. Cells were transfected with *abcB3* expression vectors or empty pcDNA3.1 plasmid using Lipofectamine 2000 (Invitrogen) according to the manufacturer's recommended protocol. Twenty-four hours after transfection, cells were counted, and re-seeded in a 6 well plate at $4.16 \times 10^4 \text{ cm}^{-2}$. The cells were used for cAMP export experiments 72 h post-transfection.

mRNA abundance analysis

D. discoideum and *D. purpureum* were developed on black nitrocellulose filters at 22°C for 0, 3, or 6 h. Cells were harvested with KK2 buffer containing 0.02 M EDTA, and 3×10^7 cells were re-suspended in 30 mL of KK2 buffer containing no cargo or 20 mM of 8-Br-cAMP. Cells were incubated in shaking suspension at 220 rpm for 2 h, harvested by centrifugation, and re-suspended in 1 mL of Trizol® (Life technologies, CA, USA). Total RNA was extracted according to the manufacturer's recommended protocol. RNA was treated with DNase as described (Benabentos et al., 2009), cDNA prepared using Superscript II reverse transcriptase as described (Parikh et al., 2010), and quantitative reverse transcriptase PCR (qRT-PCR) performed with oligonucleotides specific to each *abc* gene as described (Miranda et al., 2013). Two independent incubation experiments with three technical replicates of the qRT-PCR were performed. To determine *abc* transcript abundance in *pkaR*[−] and *pkaOE* strains, experiments and analysis were performed similarly but without 8-Br-cAMP treatment. To confirm *abcB3* knockdown, qRT-PCR was performed as above after 10 h of development.

Intracellular and extracellular cAMP measurement

These experiments were conducted as described (Van Haastert, 2006). Briefly, *D. discoideum* cells were developed on black nitrocellulose filters for 4–6 h, harvested using KK2 buffer containing 0.02 M EDTA, and re-suspended at 5×10^7 cells/mL in KK2 buffer. Using 4×10^6 cells of the suspension (80 µL), cAMP export was stimulated with 20 µL of a stimulus solution containing 50 µM 2'-deoxy-cAMP and 50 mM DTT for 5 min. Samples were centrifuged for 5 s at 14,000g and intracellular and supernatant fractions were separated. Stimulation was stopped by adding 100 µL of 3.5% (v/v) perchloric acid to fractions and stored at −20°C for radiolabeled competitive cAMP binding assays.

Human cells were washed with 2 mL of $1 \times$ phosphate buffer saline (PBS, Invitrogen) without dislodging and incubated for 30 min with 1 mL of PBS containing 0.2 mM 3-isobutyl-1-methylxanthine (IBMX) (Sigma-Aldrich), or 0.2 mM IBMX and 100 µM Forskolin (Sigma-Aldrich), or 0.2 mM IBMX, 100 µM of Forskolin, and 100 µM Probenecid. Cells were harvested, centrifuged at 600g for 5 min, and

the cells suspension was separated into supernatant (extracellular fraction) and pellet (intracellular fraction). Stimulation was terminated by adding 100 µL of 3.5% (v/v) perchloric acid to the fractions. Samples were stored at −20°C for radiolabeled competitive cAMP binding assays.

cAMP measurement was assayed as described in Van Haastert (2006). Briefly, samples were neutralized using a 50% saturated solution of KHCO₃, centrifuged, and the supernatant collected for the assay. 20 µL of tracer (0.4 µCi/mL of ³H-cAMP), 20 µL of sample or standard, and 40 µL of cAMP binding protein prepared from beef muscle (Chen et al., 2002) were combined and incubated on ice for 1.5 h. Binding reactions were stopped by addition of 40 µL activated charcoal. 50 µL of the supernatant was used to measure radioactivity via scintillation counting, and the amount of cAMP in samples was determined by comparing samples to standard amounts of cAMP as described (Van Haastert, 2006). For each experiment, samples were prepared in at least duplicates.

Acknowledgments

We thank Olga Zuchenko for help at early stages of the research and for fruitful discussions. This work was supported by the Dictyostelium Functional Genomics Program Project Grant from the National Institutes of Health (P01 HD39691). E.N. was supported by the IRACDA K12 GM084897 grant from the National Institutes of Health. The sponsors had no involvement in the study design, the data collection, analysis and interpretation, the decision to submit the work for publication or the writing of the report.

References

- Alvarez-Curto, E., et al., 2005. Evolutionary origin of cAMP-based chemoattraction in the social amoebae. *Proc. Natl. Acad. Sci. USA* 102 (18), 6385–6390.
- Anjard, C., Loomis, W.F., 2002. Evolutionary analyses of ABC transporters of *Dictyostelium discoideum*. *Eukaryot. Cell* 1 (4), 643–652.
- Anjard, C., Pinaud, S., Kay, R.R., Reymond, C.D., 1992. Overexpression of Dd PK2 protein kinase causes rapid development and affects the intracellular cAMP pathway of *Dictyostelium discoideum*. *Development* 115 (3), 785–790.
- Annilo, T., et al., 2006. Evolution of the vertebrate ABC gene family: analysis of gene birth and death. *Genomics* 88 (1), 1–11.
- Ardaillou, N., Placier, S., Wahbe, F., Ronco, P., Ardaillou, R., 1993. Release of cyclic nucleotides from the apical and basolateral poles of cultured human glomerular epithelial cells. *Exp. Nephrol.* 1 (4), 253–260.
- van Aubel, R.A., Smeets, P.H., Peters, J.G., Bindels, R.J., Russel, F.G., 2002. The MRP4/ABCC4 gene encodes a novel apical organic anion transporter in human kidney proximal tubules: putative efflux pump for urinary cAMP and cGMP. *J. Am. Soc. Nephrol.* 13 (3), 595–603.
- Barkley, D.S., 1969. Adenosine-3',5'-phosphate: identification as acrasin in a species of cellular slime mold. *Science* 165 (3898), 1133–1134.
- Benabentos, R., et al., 2009. Polymorphic members of the lag gene family mediate kin discrimination in *Dictyostelium*. *Curr. Biol.* 19 (7), 567–572.
- Cheepala, S., et al., 2013. Cyclic nucleotide compartmentalization: contributions of phosphodiesterases and ATP-binding cassette transporters. *Annu. Rev. Pharmacol. Toxicol.* 53, 231–253.
- Chen, H.L., McCauley, L.K., D'Silva, N.J., 2002. cAMP binding protein assay for widespread use in cell signaling studies. *Biotechniques* 33 (1) (66–68, 70, 72).
- Chen, Z.S., Lee, K., Kruh, G.D., 2001. Transport of cyclic nucleotides and estradiol 17-beta-D-glucuronide by multidrug resistance protein 4. Resistance to 6-mercaptopurine and 6-thioguanine. *J. Biol. Chem.* 276 (36), 33747–33754.
- Dean, M., Hamon, Y., Chimini, G., 2001. The human ATP-binding cassette (ABC) transporter superfamily. *J. Lipid Res.* 42 (7), 1007–1017.
- Dinauer, M.C., MacKay, S.A., Devreotes, P.N., 1980. Cyclic 3',5'-AMP relay in *Dictyostelium discoideum* III. The relationship of cAMP synthesis and secretion during the cAMP signaling response. *J. Cell Biol.* 86 (2), 537–544.
- Frelet, A., Klein, M., 2006. Insight in eukaryotic ABC transporter function by mutation analysis. *FEBS Lett.* 580 (4), 1064–1084.
- Graf, G.A., et al., 2003. ABCG5 and ABCG8 are obligate heterodimers for protein trafficking and biliary cholesterol excretion. *J. Biol. Chem.* 278 (48), 48275–48282.
- Hamet, P., Pang, S.C., Tremblay, J., 1989. Atrial natriuretic factor-induced egression of cyclic guanosine 3':5'-monophosphate in cultured vascular smooth muscle and endothelial cells. *J. Biol. Chem.* 264 (21), 12364–12369.

- Heasley, L.E., Brunton, L.L., 1985. Prostaglandin A1 metabolism and inhibition of cyclic AMP extrusion by avian erythrocytes. *J. Biol. Chem.* 260 (21), 11514–11519.
- Hirose, S., Benabentos, R., Ho, H.I., Kuspa, A., Shaulsky, G., 2011. Self-recognition in social amoebae is mediated by allelic pairs of tiger genes. *Science* 333 (6041), 467–470.
- Kessin, R.H., 2001. *Dictyostelium – Evolution, Cell Biology, and the development of multicellularity*. Cambridge University Press, Cambridge, UK.
- Knecht, D.A., Cohen, S.M., Loomis, W.F., Lodish, H.F., 1986. Developmental regulation of *Dictyostelium discoideum* actin gene fusions carried on low-copy and high-copy transformation vectors. *Mol. Cell. Biol.* 6 (11), 3973–3983.
- Linton, K.J., 2007. Structure and function of ABC transporters. *Physiology (Bethesda)* 22, 122–130.
- Loomis, W.F., Shaulsky, G., 2011. Developmental changes in transcriptional profiles. *Dev. Growth Differ.* 53 (4), 567–575.
- Mann, S.K., Yonemoto, W.M., Taylor, S.S., Firtel, R.A., 1992. DdPK3, which plays essential roles during *Dictyostelium* development, encodes the catalytic subunit of cAMP-dependent protein kinase. *Proc. Natl. Acad. Sci. USA* 89 (22), 10701–10705.
- Martens, H., et al., 2002. RNAi in *Dictyostelium*: the role of RNA-directed RNA polymerases and double-stranded RNase. *Mol. Biol. Cell* 13 (2), 445–453.
- Meima, M., Schaap, P., 1999. *Dictyostelium* development-socializing through cAMP. *Semin. Cell Dev. Biol.* 10 (6), 567–576.
- Miranda, E.R., et al., 2013. ABC transporters in *Dictyostelium discoideum* development. *PLoS One* 8 (8), e70040.
- Nam, E.A., Zhao, R., Cortez, D., 2011a. Analysis of mutations that dissociate G(2) and essential S phase functions of human ataxia telangiectasia-mutated and Rad3-related (ATR) protein kinase. *J. Biol. Chem.* 286 (43), 37320–37327.
- Nam, E.A., et al., 2011b. Thr-1989 phosphorylation is a marker of active ataxia telangiectasia-mutated and Rad3-related (ATR) kinase. *J. Biol. Chem.* 286 (33), 28707–28714.
- Oram, J.F., Lawn, R.M., Garvin, M.R., Wade, D.P., 2000. ABCA1 is the cAMP-inducible apolipoprotein receptor that mediates cholesterol secretion from macrophages. *J. Biol. Chem.* 275 (44), 34508–34511.
- Parikh, A., et al., 2010. Conserved developmental transcriptomes in evolutionarily divergent species. *Genome Biol.* 11 (3), R35.
- Patel, M.J., Wypij, D.M., Rose, D.A., Rimele, T.J., Wiseman, J.S., 1995. Secretion of cyclic GMP by cultured epithelial and fibroblast cell lines in response to nitric oxide. *J. Pharmacol. Exp. Ther.* 273 (1), 16–25.
- Pitt, G.S., et al., 1992. Structurally distinct and stage-specific adenylyl cyclase genes play different roles in *Dictyostelium* development. *Cell* 69 (2), 305–315.
- Pitt, G.S., Brandt, R., Lin, K.C., Devreotes, P.N., Schaap, P., 1993. Extracellular cAMP is sufficient to restore developmental gene expression and morphogenesis in *Dictyostelium* cells lacking the aggregation adenylyl cyclase (ACA). *Genes Dev.* 7 (11), 2172–2180.
- Prabhu, Y., Eichinger, L., 2006. The *Dictyostelium* repertoire of seven transmembrane domain receptors. *Eur. J. Cell Biol.* 85 (9–10), 937–946.
- Reid, G., et al., 2003. The human multidrug resistance protein MRP4 functions as a prostaglandin efflux transporter and is inhibited by nonsteroidal antiinflammatory drugs. *Proc. Natl. Acad. Sci. USA* 100 (16), 9244–9249.
- Richardson, D.L., Loomis, W.F., Kimmel, A.R., 1994. Progression of an inductive signal activates sporulation in *Dictyostelium discoideum*. *Development* 120 (10), 2891–2900.
- Riley, B.B., Barclay, S.L., 1990. Conditions that alter intracellular cAMP levels affect expression of the cAMP phosphodiesterase gene in *Dictyostelium*. *Proc. Natl. Acad. Sci. USA* 87 (12), 4746–4750.
- Saran, S., Schaap, P., 2004. Adenylyl cyclase G is activated by an intramolecular osmosensor. *Mol. Biol. Cell* 15 (3), 1479–1486.
- Saran, S., et al., 2002. cAMP signaling in *Dictyostelium*. Complexity of cAMP synthesis, degradation and detection. *J. Muscle Res. Cell Motil.* 23 (7–8), 793–802.
- Sarkadi, B., Homolya, L., Szakacs, G., Varadi, A., 2006. Human multidrug resistance ABCB and ABCG transporters: participation in a chemoinnate defense system. *Physiol. Rev.* 86 (4), 1179–1236.
- Saxe 3rd, C.L., Johnson, R., Devreotes, P.N., Kimmel, A.R., 1991. Multiple genes for cell surface cAMP receptors in *Dictyostelium discoideum*. *Dev. Genet.* 12 (1–2), 6–13.
- Schaap, P., 2011. Evolutionary crossroads in developmental biology: *Dictyostelium discoideum*. *Development* 138 (3), 387–396.
- Schaap, P., et al., 2006. Molecular phylogeny and evolution of morphology in the social amoebas. *Science* 314 (5799), 661–663.
- Shaulsky, G., Fuller, D., Loomis, W.F., 1998. A cAMP-phosphodiesterase controls PKA-dependent differentiation. *Development* 125 (4), 691–699.
- Sucgang, R., et al., 2011. Comparative genomics of the social amoebae *Dictyostelium discoideum* and *Dictyostelium purpureum*. *Genome Biol.* 12 (2), R20.
- Sucgang, R., et al., 2013. Comparative genomics of the social amoebae *Dictyostelium discoideum* and *Dictyostelium purpureum*. *Genome Biol.* 12 (2), R20.
- Sussman, M., 1987. Cultivation and synchronous morphogenesis of *Dictyostelium* under controlled experimental conditions. *Methods Cell Biol.* 28, 9–29.
- Tjornhammar, M.L., Lazaridis, G., Bartfai, T., 1983. Cyclic GMP efflux from liver slices. *J. Biol. Chem.* 258 (11), 6882–6886.
- Van Haastert, P.J., 2006. Analysis of signal transduction: formation of cAMP, cGMP, and Ins(1,4,5)P3 in vivo and in vitro. *Methods Mol. Biol.* 346, 369–392.
- van Veen, H.W., et al., 1998. A bacterial antibiotic-resistance gene that complements the human multidrug-resistance P-glycoprotein gene. *Nature* 391 (6664), 291–295.
- Venkateswaran, A., et al., 2000a. Control of cellular cholesterol efflux by the nuclear oxysterol receptor LXR alpha. *Proc. Natl. Acad. Sci. USA* 97 (22), 12097–12102.
- Venkateswaran, A., et al., 2000b. Human white/murine ABC8 mRNA levels are highly induced in lipid-loaded macrophages. A transcriptional role for specific oxysterols. *J. Biol. Chem.* 275 (19), 14700–14707.
- Virdy, K.J., et al., 1999. High cAMP in spores of *Dictyostelium discoideum*: association with spore dormancy and inhibition of germination. *Microbiology* 145 (Pt 8), 1883–1890.
- Wang, B., Kuspa, A., 1997. *Dictyostelium* development in the absence of cAMP. *Science* 277 (5323), 251–254.
- Wielinga, P.R., et al., 2003. Characterization of the MRP4- and MRP5-mediated transport of cyclic nucleotides from intact cells. *J. Biol. Chem.* 278 (20), 17664–17671.
- Zhou, X., et al., 2008. Genetic deletion of low density lipoprotein receptor impairs sterol-induced mouse macrophage ABCA1 expression. A new SREBP1-dependent mechanism. *J. Biol. Chem.* 283 (4), 2129–2138.
- van de Water, F.M., Masereeuw, R., Russel, F.G., 2005. Function and regulation of multidrug resistance proteins (MRPs) in the renal elimination of organic anions. *Drug Metab. Rev.* 37 (3), 443–471.